department	*status*	*age*	*salary*
sales	senior	31...35	46K...50K
sales	junior	26...30	26K...30K
sales	junior	31...35	31K...35K
systems	junior	21...25	46K...50K
systems	senior	31...35	66K...70K
systems	junior	26...30	46K...50K
systems	senior	41...45	66K...70K
marketing	senior	36...40	46K...50K
marketing	junior	31...35	41K...45K
secretary	senior	46...50	36K...40K
secretary	junior	26...30	26K...30K

Train a naïve Bayesian classifier as the status attribute is the class label and all the remaining attributes are regarded as input. Once you have your naïve Bayesian classifier, test the following unseen instances: □1. (sales,31…35, 66K…70K).

We treat **status** as the class label:

- Classes: **Senior**, **Junior**
- Features: **Department, Age, Salary**

Dataset size = **11 records**

---

1. Prior Probabilities

$$P(Senior) = \frac{5}{11}, \quad P(Junior) = \frac{6}{11}$$

Counts from the table:

Status	Count
Senior	5
Junior	6

---

2. Conditional Probabilities

For Senior (5 records)

Department

Department	Count	Probability
sales	1	1/5
systems	2	2/5
marketing	1	1/5
secretary	1	1/5

Age

Age	Count	Probability
31–35	2	2/5
41–45	1	1/5
36–40	1	1/5
46–50	1	1/5

Salary

Salary	Count	Probability
46–50K	2	2/5
66–70K	2	2/5
36–40K	1	1/5

For Junior (6 records)

Department

Department	Count	Probability
sales	2	2/6
systems	2	2/6
marketing	1	1/6
secretary	1	1/6

Age

Age	Count	Probability
26–30	3	3/6
31–35	2	2/6
21–25	1	1/6

Salary

Salary	Count	Probability
26–30K	2	2/6
31–35K	1	1/6
46–50K	2	2/6
41–45K	1	1/6

Note: **No Junior record with salary 66–70K**

---

3. Test Instance

Instance:

```
(sales, 31–35, 66–70K)
```

---

4. Naïve Bayes Calculation

For Senior

$$P(Senior|X) \propto P(Senior) \times P(sales|Senior) \times P(31\text{-}35|Senior) \times P(66\text{-}70|Senior$$

$$= \frac{5}{11} \times \frac{1}{5} \times \frac{2}{5} \times \frac{2}{5}$$

$$= \frac{20}{1375} \approx 0.0145$$

For Junior

$$P(Junior|X) \propto P(Junior) \times P(sales|Junior) \times P(31\text{-}35|Junior) \times P(66\text{-}70|Junio$$

$$P(66\text{-}70|Junior) = 0$$

$$P(Junior|X) = 0$$

---

5. Final Classification

Class	Probability
Senior	0.0145
Junior	0

✅ **Predicted Class = Senior**

✔ **Final Answer:**

The unseen instance **(sales, 31–35, 66K–70K)** is classified as **Senior**.

If you want, I can also show **how this would change using Laplace smoothing (which professors often expect in exams)**.